

Quantitative evaluation of activation maps for weakly-supervised lung nodule segmentation

March 14, 2025

Rachida Saroui

ENS Paris-Saclay

`rachida.Saroui@ens-paris-saclay.fr`

Ilias Rami

ENS Paris-Saclay

`ilias.rami@ens-paris-saclay.fr`

1 Paper Summary

Behrendt *et al.* [1] address the challenge of weakly-supervised lung nodule segmentation in chest X-rays using deep learning. Accurate nodule detection is crucial for early lung cancer diagnosis, but manual review is time-consuming and error-prone [2]. Deep learning models offer promising support in this context [3]; however, their limited explainability often reduces clinical trust. To bridge this gap, the study explores whether activation maps from classification models can effectively localize nodules without dedicated segmentation training. To achieve this, the authors evaluate activation maps from three classifiers: DenseNet-121, DeiT, and PatchConvNet. Each model is first trained for nodule classification, after which activation maps are generated using Grad-CAM [4]. In addition to Grad-CAM, the transformer models provide internal self-attention maps, adding an extra dimension for analysis. Finally, these activation and attention maps are compared to ground truth nodule locations to assess their effectiveness in delivering explainable localization.

Positioning with Respect to the State of the Art

Researchers have developed methods like Class Activation Mapping (CAM)[5] and Gradient-weighted Class Activation Mapping (Grad-CAM) [4] to highlight regions of focus using image-level labels. However, [6] [8][7] have raised concerns about their reliability in medical imaging, as saliency maps may not consistently correspond to true disease locations. To improve weakly-supervised localization, strategies such as integrating attention mechanisms into CNNs and employing multiple instance learning have been proposed[10] [9]. Notably, ViTs[11] offer interpretability by providing attention maps that indicate the model's focus areas, and recent evidence suggests they can match or surpass CNN performance in medical imaging tasks [12].

Main Contributions

This study compares CNN (DenseNet-121), transformer (DeiT), and hybrid (PatchConvNet) models under identical conditions, revealing differences in feature learning. Grad-CAM and transformer attention maps are evaluated, with the hybrid model's self-attention achieving the best, though still low, localization accuracy. The findings highlight that high classification accuracy does not ensure effective localization, as activation and attention maps often focus on irrelevant features.

Methodology

The study followed a two-stage process involving model training and post-processing for lung nodule detection, using Node21 dataset, and segmentation. Initially, DenseNet-121, DeiT, and PatchConvNet models were trained for the classification task. Subsequently, Grad-CAM was applied to highlight activation maps, and for the transformer-based model, self-attention matrices were extracted to visualize the regions the model focused on during prediction. Classification performance was assessed using AUROC and AUPRC metrics, while segmentation performance was evaluated via pixel-wise AUPRC by comparing activation maps against ground-truth nodule masks.

Results

The study revealed that while all three models—DenseNet-121, DeiT, and PatchConvNet—effectively detected lung nodules for Node21 dataset, their ability to localize these nodules was limited. Despite high classification accuracy, the models’ activation maps poorly corresponded to actual nodule locations, indicating reliance on contextual cues rather than direct identification. This underscores that high classification performance does not necessarily equate to accurate localization in weakly-supervised settings.

2 Critical Assessment

Strengths

A key strength of the work is its comprehensive comparative approach. The authors evaluate multiple state-of-the-art model architectures, including a CNN, a ViT, and a hybrid model. Another strength is the rigorous experimental design: the study leverages a sizable dataset of 4,882 chest radiographs assembled from multiple public sources, with radiologist-provided annotations for nodule locations. The paper’s quantitative evaluation of localization using attention maps is a notable contribution, as prior works often only qualitatively assess saliency maps for CXR’s images. The paper does well to discuss possible reasons, noting that the CNN-based maps often focus on irrelevant but correlated features (e.g., lung edges or heart contour) instead of the nodules, whereas the transformer’s attention maps performed slightly better in highlighting true nodule regions. The authors situate their work in context by citing relevant literature on explainable AI and medical image analysis. The work’s primary contribution is thus a well-documented empirical study that can guide future research toward more explainable and trustworthy models.

Weaknesses

While the study is thorough, its scope is primarily an empirical evaluation, which may limit the novelty of the contributions. The paper does not propose a new method for improving weakly-supervised localization, but rather evaluates existing mechanisms. As a result, some findings, though important, confirm what the community might suspect. Prior research has noted that saliency maps can be misleading or misaligned with true pathology locations and [7] tested several saliency methods but only for DensNet, so the contribution here is somewhat incremental in demonstrating this specifically for lung nodules. Another shortcoming is that the study explores a relatively narrow set of visualization techniques. The authors compared a standard CNN activation map approach and the internal attention of a ViT, but other interpretability or weakly-supervised localization methods were not evaluated. Moreover, the evaluation of segmentation performance uses radiologist-annotated bounding boxes as ground truth for nodules. This choice, while practical, means the ground truth is coarse; a small activation region that correctly marks the nodule might still score a low AUPRC against a large bounding box. The paper could have strengthened its analysis by reporting additional localization metrics to complement the AUPRC. Another potential concern is the extremely high classification accuracy achieved. Such performance raises the question of whether models might be overfitting or leveraging dataset-specific biases.

Recommendations for Improvement

To enhance this research, broadening the range of explainability techniques, such as Grad-CAM++ or attention rollout, may facilitate improved localization or provide insights if results remain unchanged. Optimizing the threshold for activation maps, potentially using a validation set, could refine the generation of binary masks, thereby enhancing alignment with known nodules. To mitigate reliance on spurious features, an ablative analysis—such as occluding key regions—would elucidate whether models depend on nodule-specific characteristics. Finally, exploring detection frameworks may be beneficial, as a two-stage detector or localization network trained with bounding boxes could outperform conventional classification models.

References

- [1] F. Behrendt *et al.*, “Quantitative evaluation of activation maps for weakly-supervised lung nodule segmentation,” in *Proc. SPIE Medical Imaging: Computer-Aided Diagnosis*, vol. 12927, 129272P, 2024.
- [2] A. P. Brady, “Error and discrepancy in radiology: inevitable or avoidable?,” *Insights into Imaging*, vol. 8, pp. 171–182, 2017.
- [3] E. Çallı *et al.*, “Deep learning for chest X-ray analysis: A survey,” *Medical Image Analysis*, vol. 72, 102125, 2021.
- [4] R. R. Selvaraju *et al.*, “Grad-CAM: Visual explanations from deep networks via gradient-based localization,” in *Proc. IEEE Int. Conf. on Computer Vision (ICCV)*, pp. 618–626, 2017.
- [5] B. Zhou *et al.*, “Learning deep features for discriminative localization,” in *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, pp. 2921–2929, 2016.
- [6] C. Ozer and I. Oksuz, “Explainable image quality analysis of chest X-rays,” in *Proc. Medical Imaging with Deep Learning (MIDL)*, 2021.
- [7] J. Zhang *et al.*, “Overlooked trustworthiness of saliency maps,” in *Proc. MICCAI, Lect. Notes Comput. Sci. (LNCS)* 13431, pp. 451–461, 2022.
- [8] H. Kang *et al.*, “Towards a quantitative analysis of class activation mapping for deep learning-based CAD,” in *Proc. SPIE Medical Imaging: Image Perception, Observer Performance, and Tech. Assessment*, vol. 11599, pp. 119–131, 2021.
- [9] E. Pesce *et al.*, “Learning to detect chest radiographs containing pulmonary lesions using visual attention networks,” *Medical Image Analysis*, vol. 53, pp. 26–38, 2019.
- [10] A. Joshi *et al.*, “Explainable disease classification via weakly-supervised segmentation,” in *Proc. MICCAI Workshop on Interpretable and Annotation-Efficient Learning for Medical Image Computing*, pp. 54–62, 2020.
- [11] A. Dosovitskiy *et al.*, “An image is worth 16x16 words: Transformers for image recognition at scale,” in *Proc. Int. Conf. on Learning Representations (ICLR)*, 2021.
- [12] F. Behrendt *et al.*, “Data-efficient vision transformers for multi-label disease classification on chest radiographs,” *Current Directions in Biomedical Engineering*, vol. 8, no. 1, pp. 34–37, 2022.